Dhiraj Shirse

+91 9373336927 | dhirajshirse2006@gmail.com | dhiraj-shirse | Dhiraj201226

Education

Veermata Jijabai Technological Institute (VJTI), Mumbai Sep 24 - Present
B.Tech in Computer Engineering | CGPA: 8.75

Late Yashwantrao Chavan Junior College, Latur Aug 22 - Mar 24
JEE Advanced: AIR 11461 | MHT-CET: 99.93% (Rank 179)| JEE Main: 97.8% | HSC: 85.5%

Relevant Coursework

- Data Structures & Algorithms
- Object Oriented Programming
- Operating Systems
- Software Engineering
- Database Management Systems
- Automata Theory

Projects

FinIntel – AI Financial Stock Advisor | *React, FastAPI, ChromaDB* May 26 – Jun 26
- Built a **RAG-powered** financial advisory platform using **React**, **FastAPI**, **ChromaDB**, and multiple LLMs to generate explainable investment insights from financial statements and company documents.
- Designed a **two-tier retrieval architecture** that first searches embedded financial documents from **ChromaDB**, then falls back to fundamental company data retrieval through **y-finance**, implemented caching to reduce redundant API calls.
- Developed a deterministic recommendation engine that evaluates companies across **Growth, Profitability, and Financial Health** dimensions and computes confidence-based **BUY/HOLD/SELL** recommendations, integrated with a multi-LLM routing layer supporting **Gemini, OpenAI** and **Ollama**.

Fake My Voice – Neural Voice Cloning System | *PyTorch, Librosa, Torchaudio* Jul 25 - Oct 25
- Built a multi-speaker voice cloning pipeline using **Tacotron2** for text-to-mel synthesis, **WaveGlow** for high-fidelity speech generation, and a **GE2E** speaker encoder with **100+ speakers** to learn discriminative speaker embeddings.
- Trained deep learning models on the **24-hour LJSpeech** dataset and multi-speaker datasets including **VCTK**, achieving **92%** speaker verification accuracy.
- Implemented text-to-phoneme conversion, mel-spectrogram generation, silence trimming, and audio denoising to improve speech naturalness and output quality. Cloned the voice with just **5 sec** audio input and text.

NeuralEye – AI-Generated Content Detection Platform | *Python,PyTorch, React, FastAPI, OpenCV* Mar 26 - Apr 26
- Built and deployed an AI-generated content detection platform using **React**, **FastAPI**, **PyTorch** for real-time image and video authenticity analysis.
- Trained and evaluated a **ResNet50**-based classifier on the **120K-image CIFAKE** dataset, achieving **98%** classification accuracy while supporting real-time predictions with confidence scoring.
- Implemented a video-analysis pipeline that extracts frames at **4-second** intervals, performs frame-level inference, and aggregates prediction scores to generate a final authenticity assessment.

Technical Skills

Languages:C++, Python, JavaScript, HTML, CSS
Frameworks & Libraries: React, PyTorch, FastAPI, NumPy
Tools & Databases: Git, GitHub, VS Code, Linux, SQLite
Concepts: RAG, Machine Learning, Neural Networks, Computer Vision, NLP

Achievements

- Reached the **Pre-Final Stage of SIH 2025** by developing a full-stack application that enabled ASHA health workers to record, manage, and synchronize healthcare data efficiently in low-connectivity regions.
- Completed Hacktoberfest 2025 with **6 pull requests** merged into open-source repositories.
- Led Red Phoenix to overall victory in Technovanza's inter-group competition by winning **VRC** – the highest point-weighted technical event.

Leadership / Extracurricular

Project X Jul 25 – Present
Core Member *VJTI*
- Member of **ProjectX**, a student-led technical club at VJTI focused on peer learning and mentorship.
- Conducted workshops and contributed to a structured mentorship program to teach and guide junior students.

Class Representative – SY B.Tech Computer Engineering, VJTI Mumbai Jul 25 – Present
- Represent 80+ students to faculty and administration; coordinate scheduling, communications, and event logistics.
- Collected student feedback and coordinated with faculty on academic and administrative matters, while managing class-wide communication of important updates and deadlines.

Department Head – Technical Committee(Technovanza 2025-26) Sep 25 – Apr 26
- Served as Department Head in the Technical Committee of Technovanza 2025–26, coordinating volunteers and event operations for a technology festival attended by 10,000+ visitors and industry exhibitors.